Generated Samples Gallery

IE 7615 Deep Learning for AI · Project 2, Milestone 2 · Group 02: Abdellah Faleh, Jin-Woo Hong, Samuel Tong, Tristan Lyons · Notebook *05_conditional_generation_sweeps.ipynb*

All images: Stable Diffusion 1.5 conditioned on **our own CLIP ViT-L/14 text embeddings**, 512×512, safety filter on (0 of 202 images blocked). The number under each image is its **CLIP score** (0–100, match with the caption). Unless the page says otherwise: PNDM sampler, 30 steps, guidance scale 7.5, scaled-linear noise schedule, seed = 42 + prompt id. Every image, prompt, seed and setting is listed in *m2_generation/results.csv*.

A. Baseline conditional outputs (15 test prompts, 2 seeds each)

Seeds 42 + id (the same images as Milestone 1, pixel difference 0.0) and 1042 + id. Mean CLIP score 27.3, 5.0 s per image.

#	Prompt (test split)	Mean CLIP
0	A tan dog running through the grass with the background blurry	27.1
1	A dog with golden fur is chest deep in water	30.7
2	A man in a number ten jersey kicks a soccer ball while another team member watches	24.8
3	A person jumping their bike in the woods in front of trees with no leaves	25.1
4	Men and women in bathing suits are standing at poolside while another swimmer is in the water	26.3
5	Boy heads into water big tents and hills in background in rural area	24.5
6	Five people on bikes in traffic with man watching from the side of the road	24.6
7	People are walking on the slippery road in winter and there are two dogs: black and yellow	29.4
8	A couple and their child pose for a picture in front of a Christmas tree	25.7
9	Three dogs who are brown white and black (respectively) play with one another in a dirt field	26.8
10	Party goers at a rave dancing by a table with glow sticks on it	29.3
11	Three women and a man sitting at a table with a large pitcher of beer	30.0
12	A group of people ride a blue roller coaster on its way down from a turn	30.9
13	A group of six children stands smiling and laughing in front of a wall	25.6
14	One man in the back of a truck working on a farm with animals in the background	29.2

Table 1. The 15 baseline prompts.

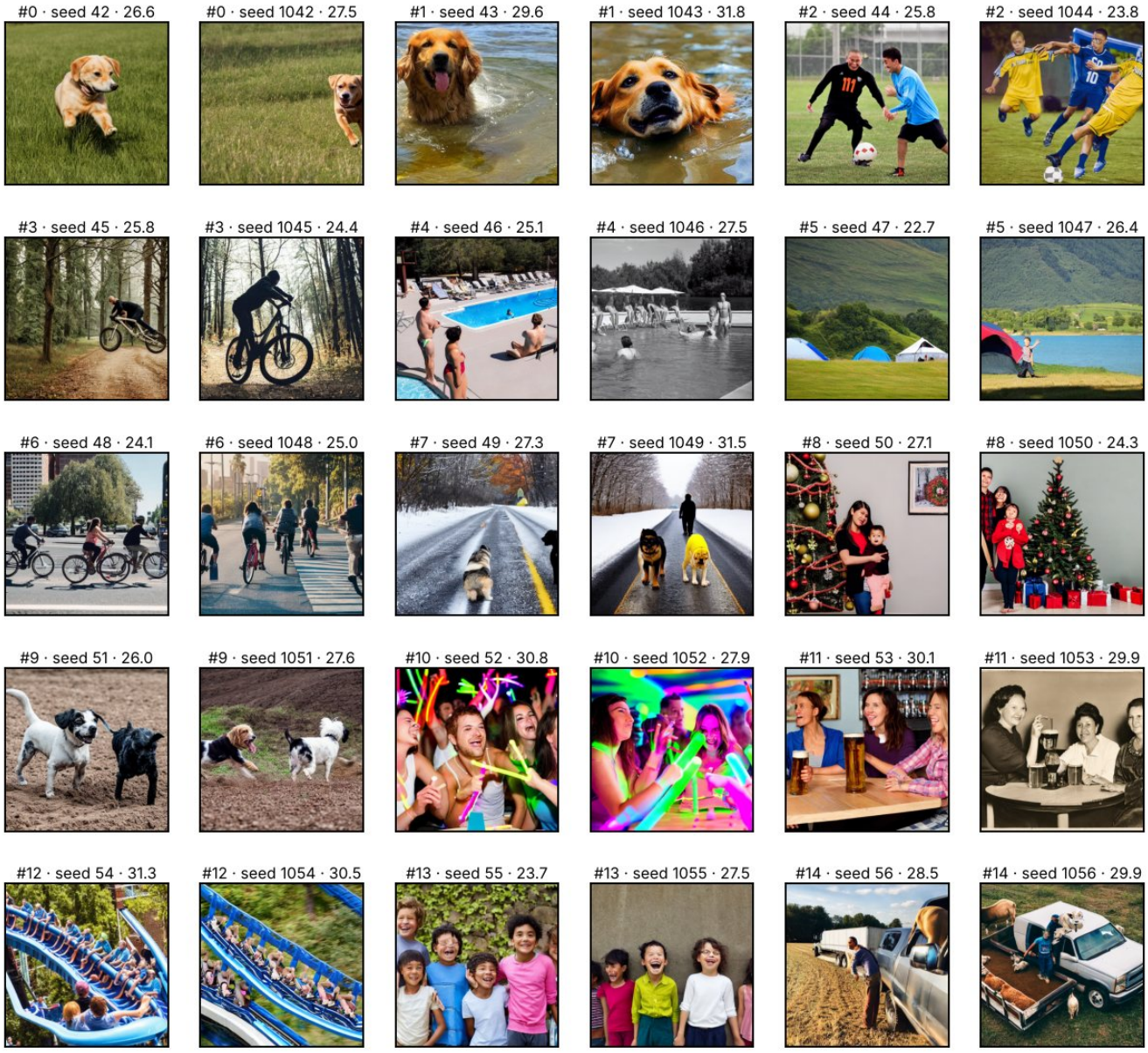


Figure 1. Baseline gallery: each prompt with seed 42 + id (left) and 1042 + id (right). Title: prompt #, seed, CLIP score.

Observations: single animals and one-idea scenes (dogs #0, #1; rave #10; roller coaster #12) look clear and match well. The Milestone 1 mistakes remain: counting (#6 five people, #9 three dogs, #11 three women and a man), the jersey number (#2 shows "111"), negation (#3 "no leaves") and missing subjects in long captions (#5).

B. Guidance-scale comparison

Same 4 prompts and seeds; only the guidance scale changes (1, 3, 5, 7.5, 10, 15). The measures below average 3 seeds.



Figure 2. Guidance scale sweep (seed 42 + id). Row labels: prompt # and short name.

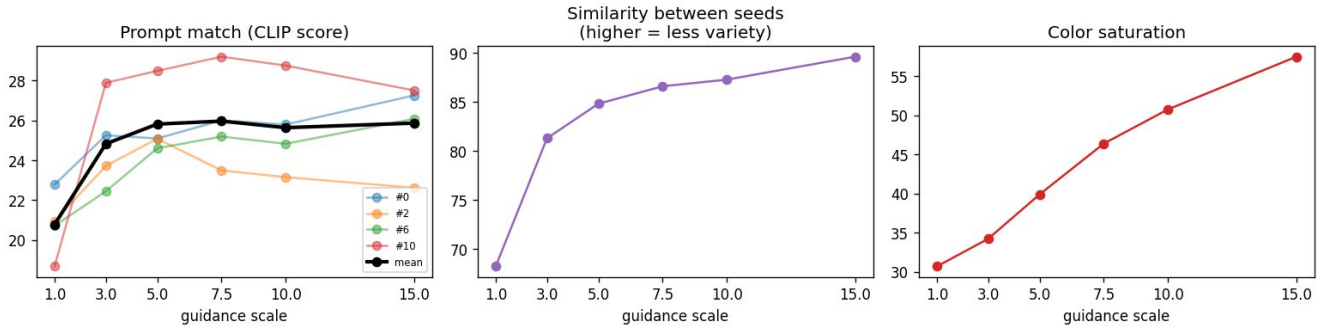


Figure 3. Mean CLIP score (left), similarity between the 3 seeds (middle, higher = less variety) and color saturation (right) against guidance scale.

Scale 1 mostly ignores the caption (the dog hides in tall grass, the rave is a collage). **3–7.5** is where the match improves. **10–15** adds no match but stronger colors and less variety.

C. Sampler × number of steps

Five samplers, 10–50 steps, guidance 7.5, seed 42 + id. Under each image: CLIP score and seconds per image on a T4.



Figure 4. #0 A tan dog running through the grass with the background blurry.



Figure 5. #2 A man in a number ten jersey kicks a soccer ball while another team member watches.



Figure 6. #6 Five people on bikes in traffic with man watching from the side of the road.



Figure 7. #10 Party goers at a rave dancing by a table with glow sticks on it.

Observations: DDPM gives a different picture at every step count (it adds new noise each step). PNDM at 10 steps is broken and DDIM at 10 steps is blurry. Euler and DPM-Solver++ look finished by 20 steps.

D. Noise-schedule comparison

DDIM with 30 steps under three beta schedules (scaled linear is the one the model was trained with), and DPM-Solver++ with 20 steps with default or Karras noise levels.

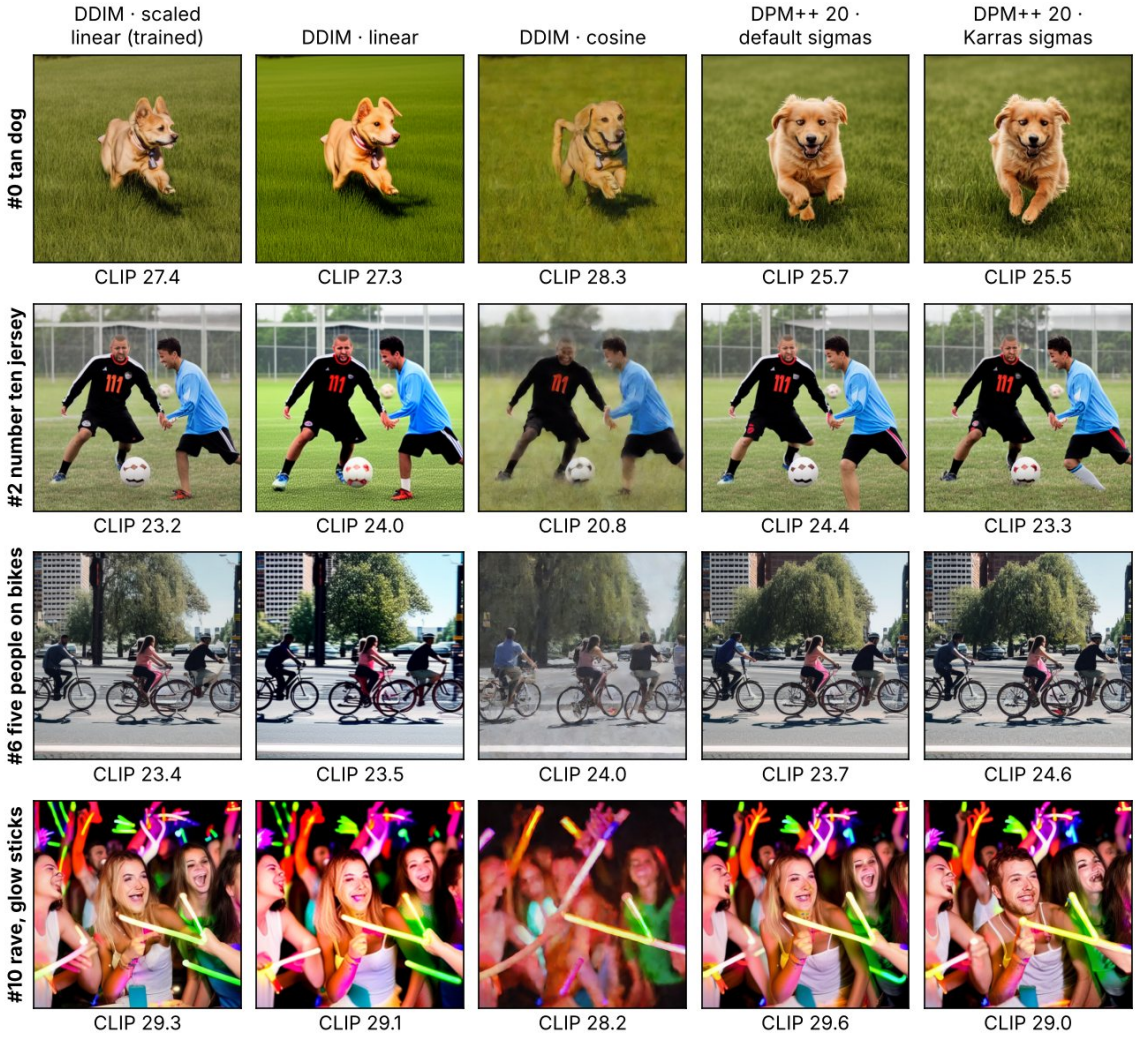


Figure 8. Noise schedule (seed 42 + id).

Observations: linear gives the same scenes with stronger, greener colors; cosine gives blurry, washed-out images (it does not match training); default and Karras sigmas look almost the same.

E. Text-encoder comparison and conditioning check (notebook 04)

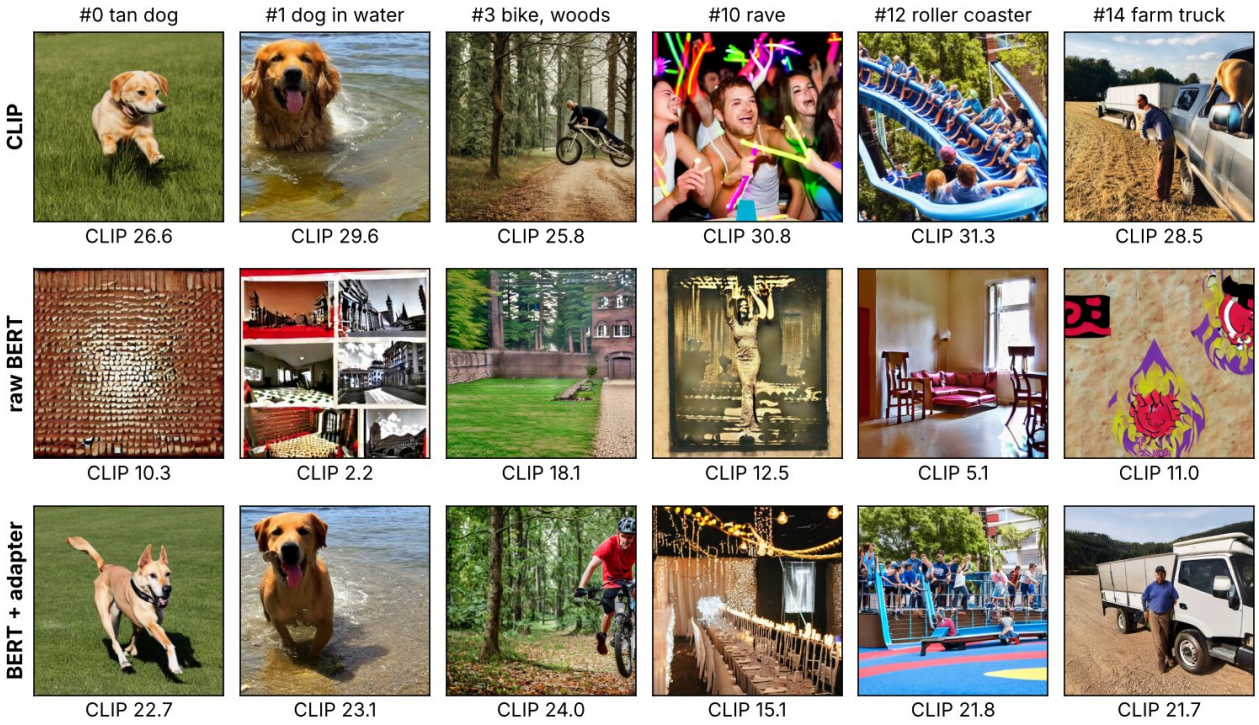


Figure 9. CLIP (top), raw BERT without adapter (middle) and BERT + our trained adapter (bottom). Guidance 7.5, PNDM 30 steps, seed 42 + id.



Figure 10. Same seed (42) for every image; only the caption changes. Last image: empty prompt. CLIP matched 6 of 6 images to their own caption.